Benchmark Results

1 2x2 Benchmark

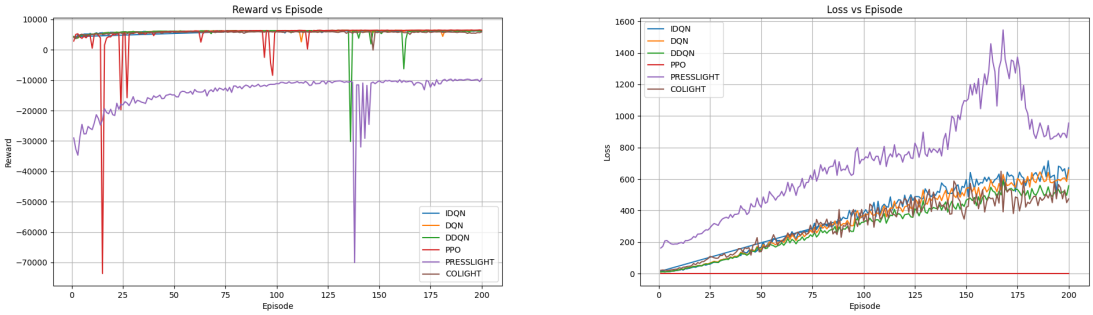


Figure 1: 2x2 Reward and Loss Comparison

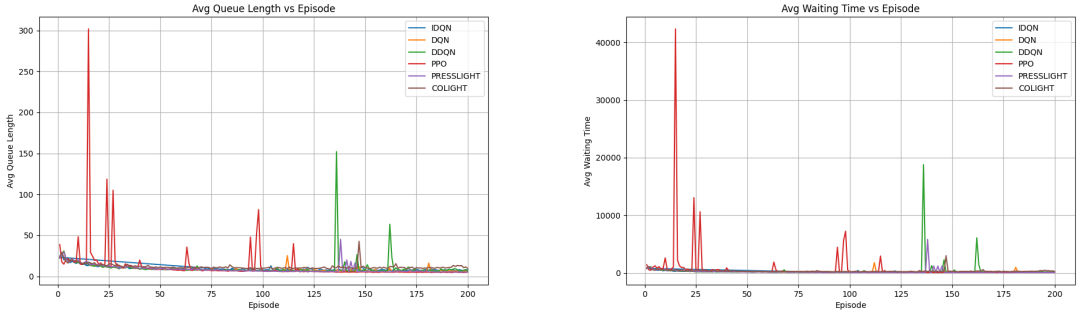


Figure 2: 2x2 Queue Length and Waiting Time Comparison

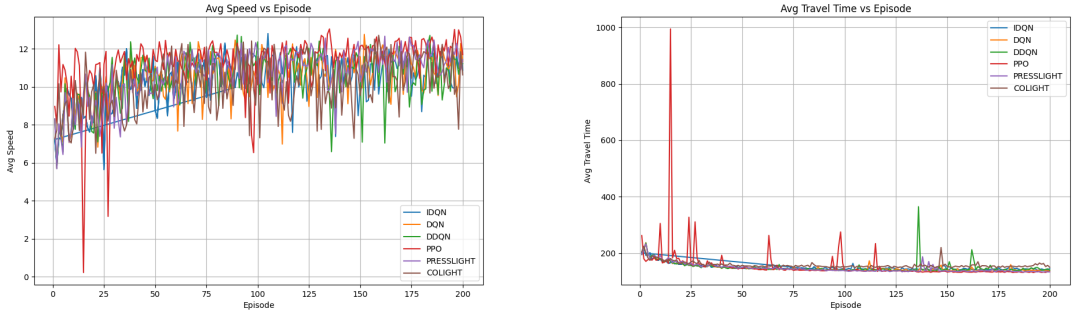


Figure 3: 2x2 Speed and Travel Time Comparison

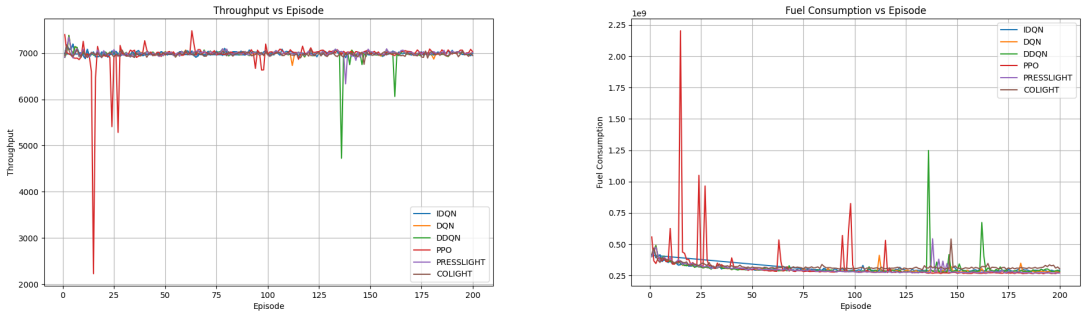


Figure 4: 2x2 Throughput and Fuel Consumption Comparison

2 4x4 Benchmark

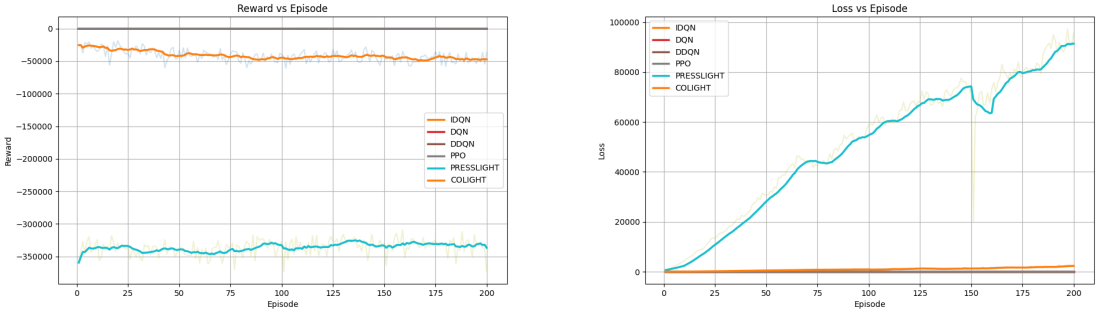


Figure 5: 4x4 Reward and Loss Comparison

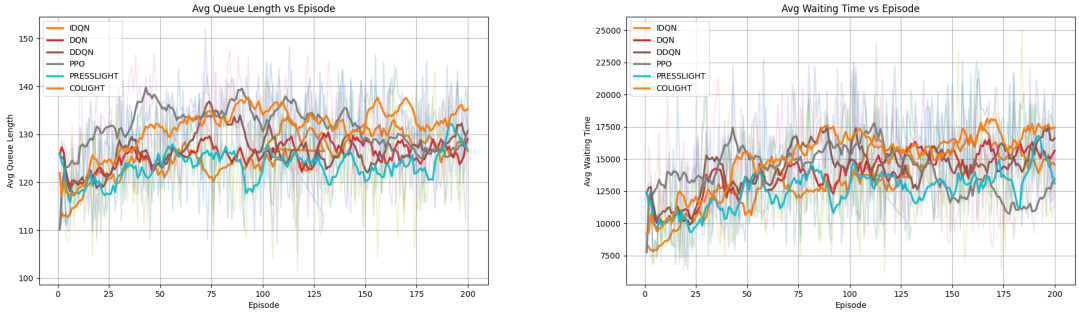


Figure 6: 4x4 Queue Length and Waiting Time Comparison

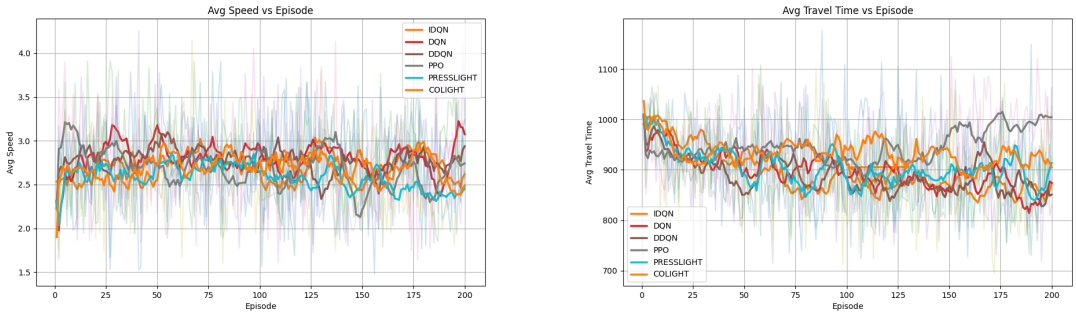


Figure 7: 4x4 Speed and Travel Time Comparison

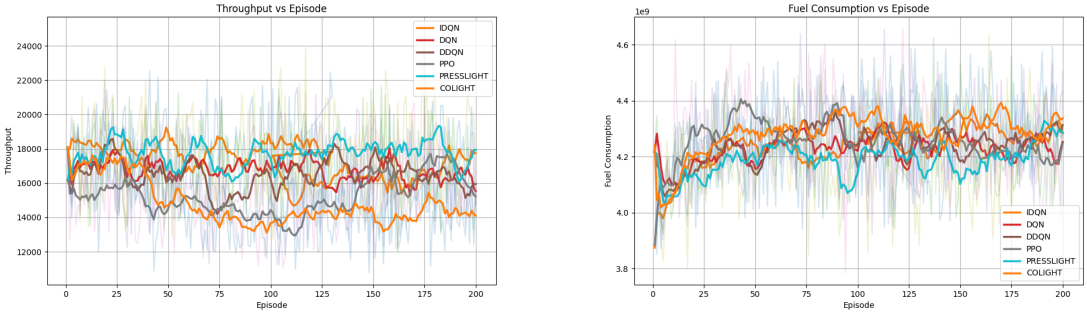


Figure 8: 4x4 Throughput and Fuel Consumption Comparison

3 6x6 Benchmark

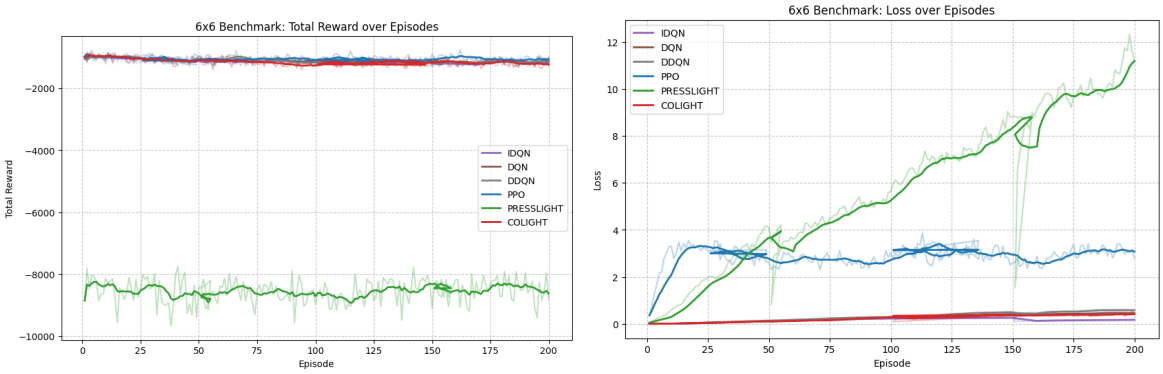


Figure 9: 6x6 Reward and Loss Comparison

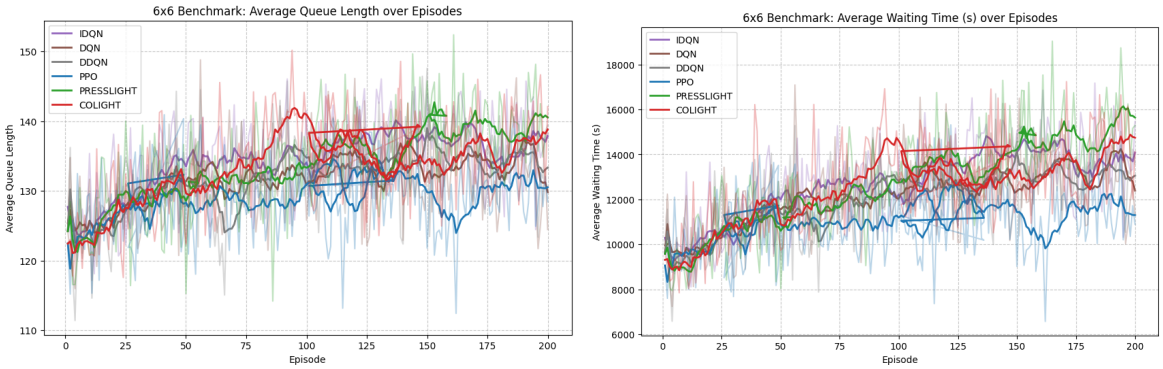


Figure 10: 6x6 Queue Length and Waiting Time Comparison

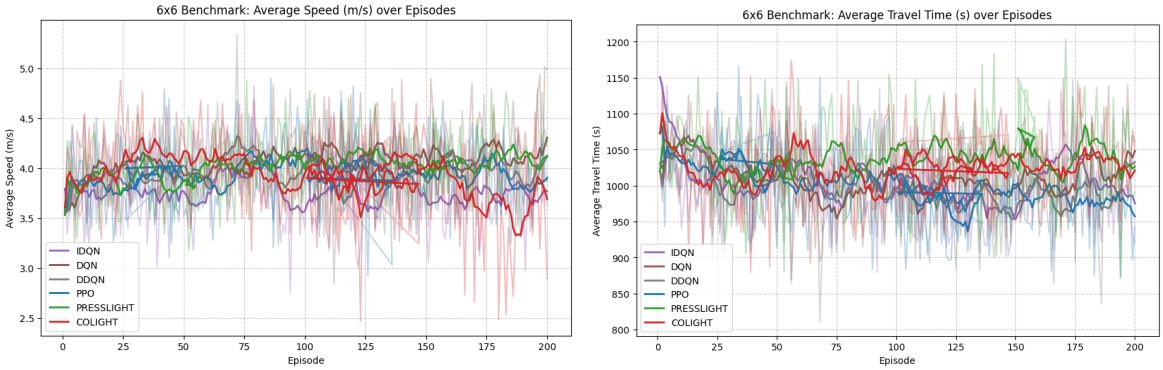


Figure 11: 6x6 Speed and Travel Time Comparison

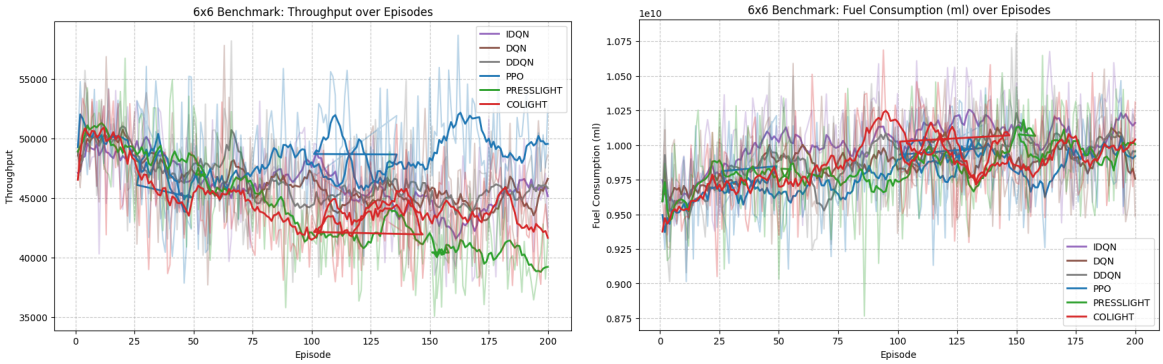


Figure 12: 6x6 Throughput and Fuel Consumption Comparison

4 PINN with RL Algorithms: 6x6 Benchmark

4.1 PPO vs PINN

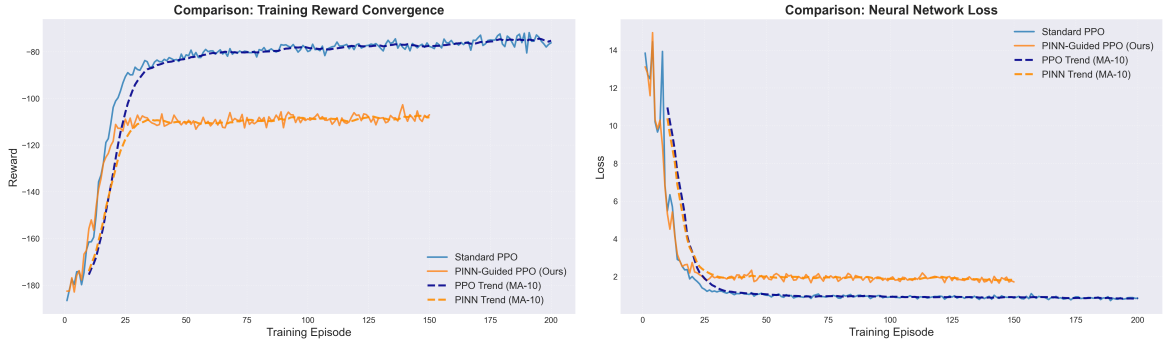


Figure 13: 6x6 PPO vs PINN Reward and Loss

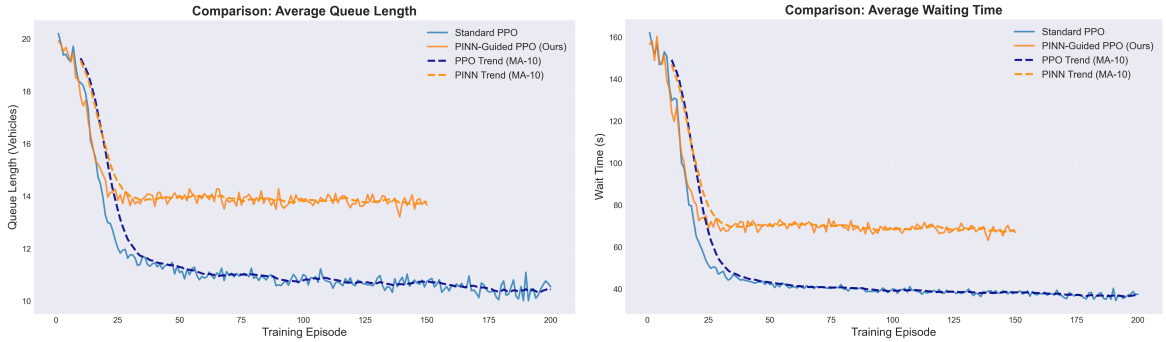


Figure 14: 6x6 PPO vs PINN Queue Length and Waiting Time

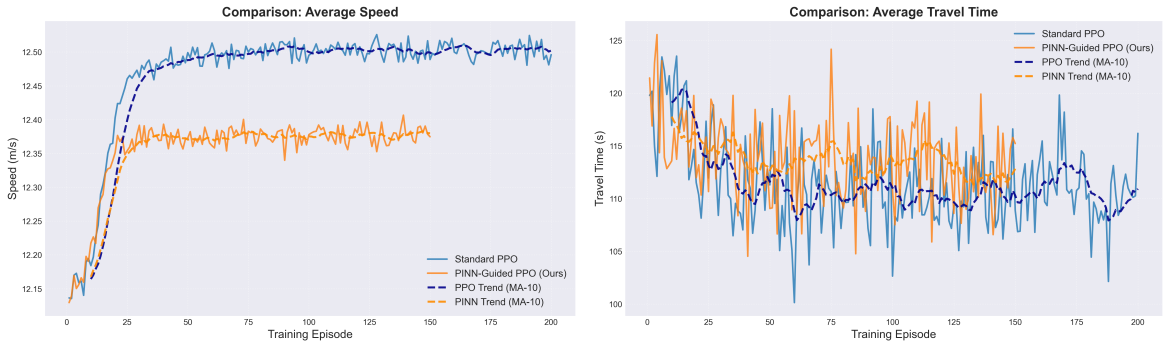


Figure 15: 6x6 PPO vs PINN Speed and Travel Time

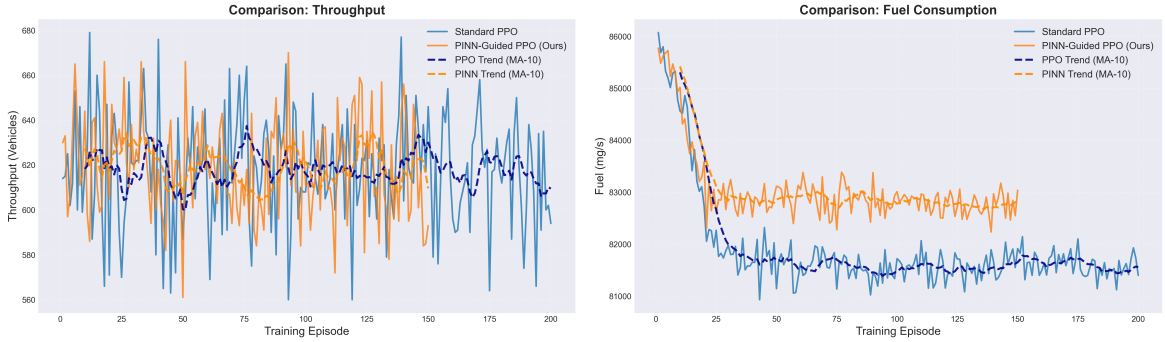


Figure 16: 6x6 PPO vs PINN Throughput and Fuel Consumption

4.2 DQN vs PINN

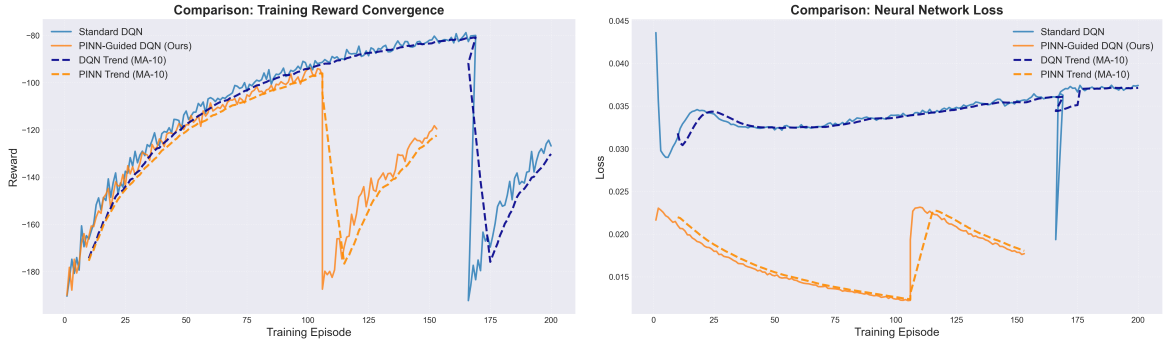


Figure 17: 6x6 DQN vs PINN Reward and Loss

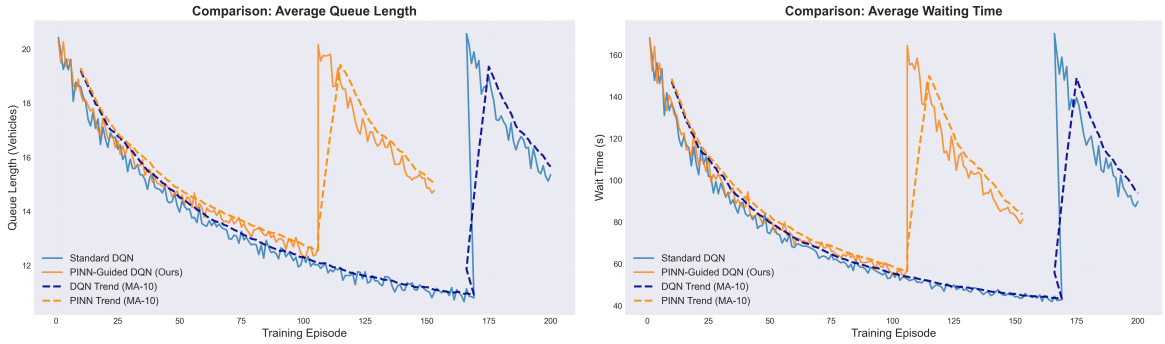


Figure 18: 6x6 DQN vs PINN Queue Length and Waiting Time

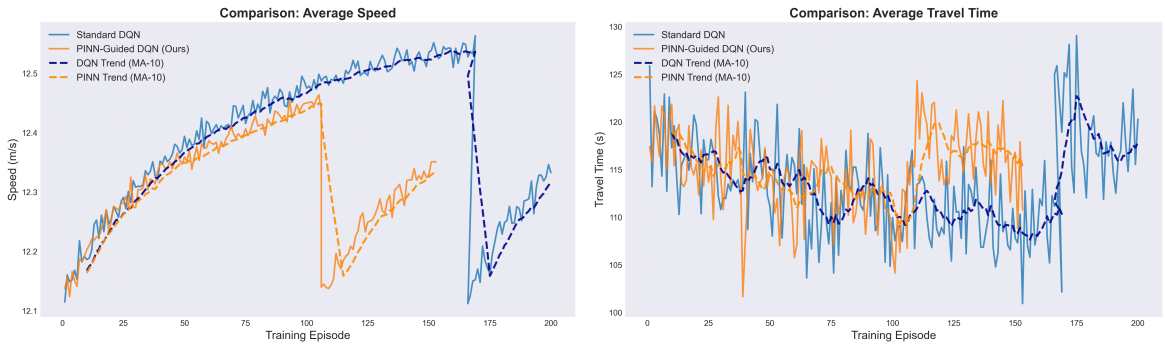


Figure 19: 6x6 DQN vs PINN Speed and Travel Time

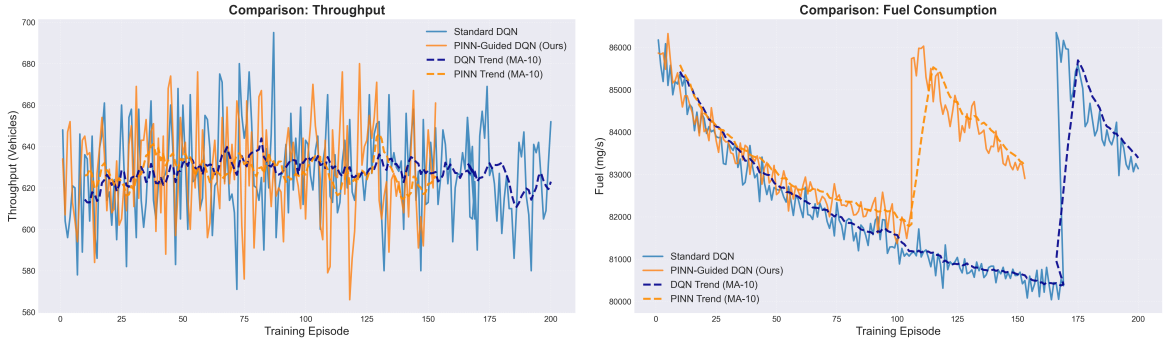


Figure 20: 6x6 DQN vs PINN Throughput and Fuel Consumption

4.3 DDQN vs PINN

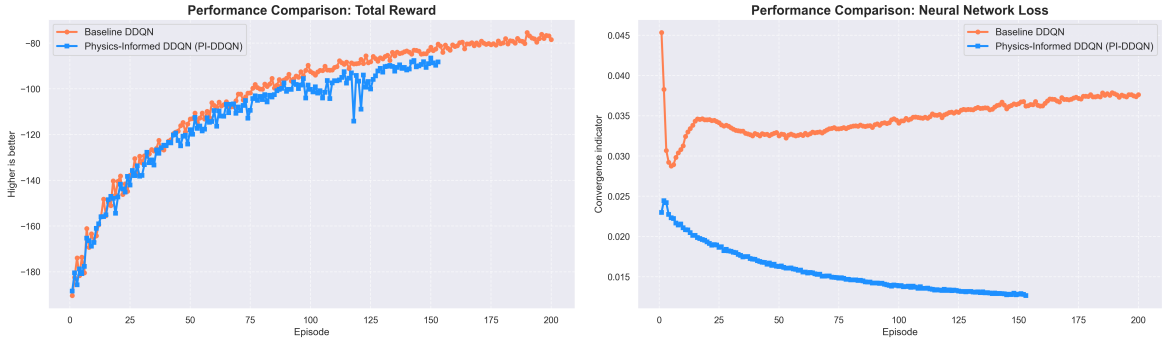


Figure 21: 6x6 DDQN vs PINN Reward and Loss

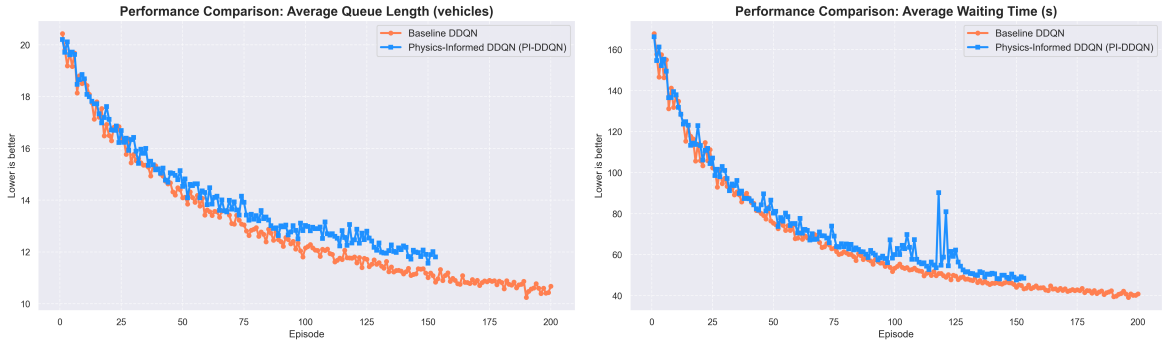


Figure 22: 6x6 DDQN vs PINN Queue Length and Waiting Time

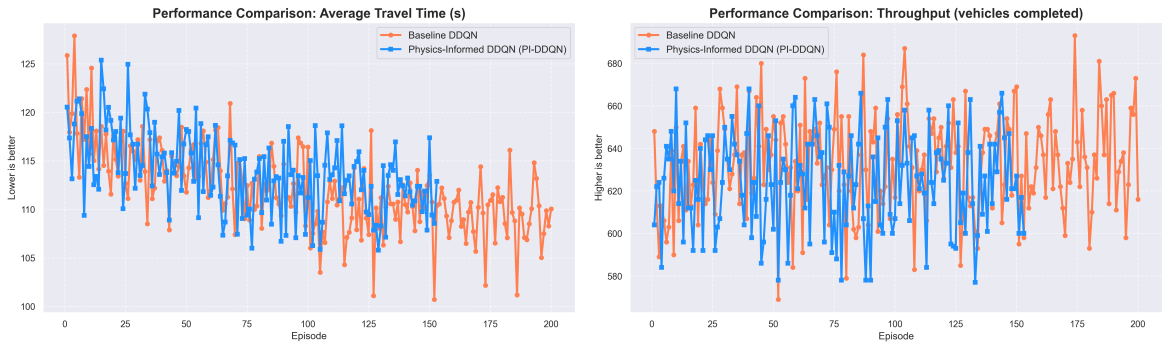


Figure 23: 6x6 DDQN vs PINN Travel Time and Throughput

5 PINN with RL Algorithms: 10x10 Benchmark

5.1 DDQN vs PINN

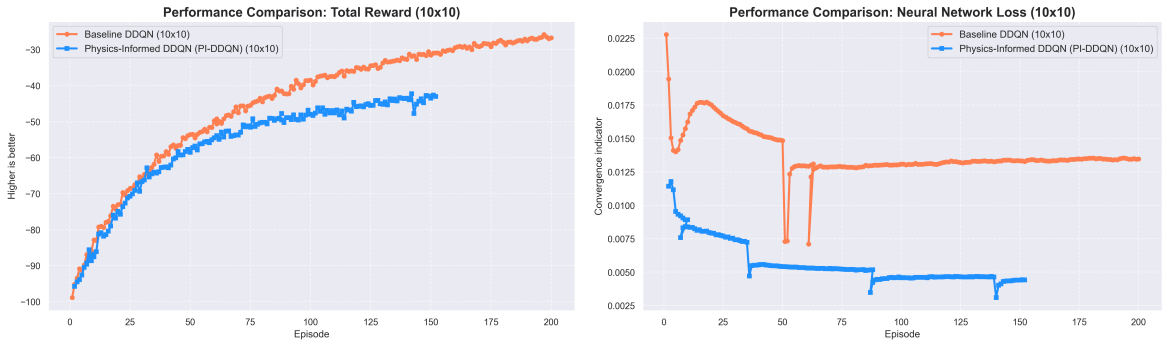


Figure 24: 10x10 DDQN vs PINN Reward and Loss

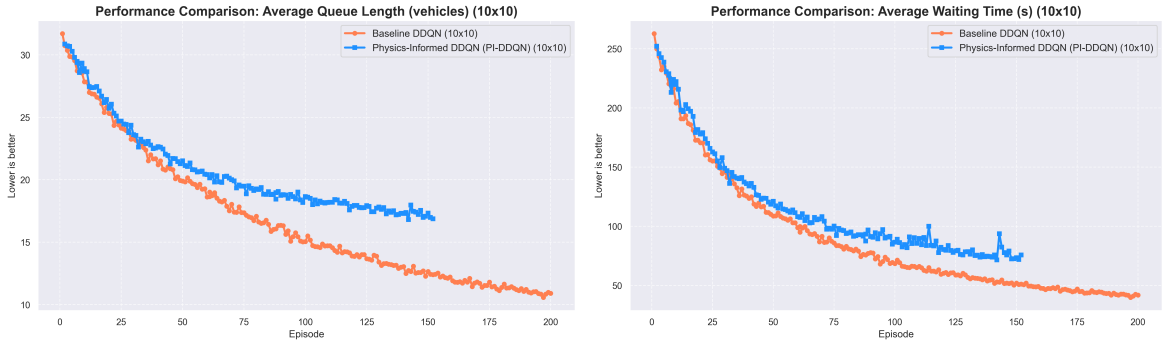


Figure 25: 10x10 DDQN vs PINN Queue Length and Waiting Time

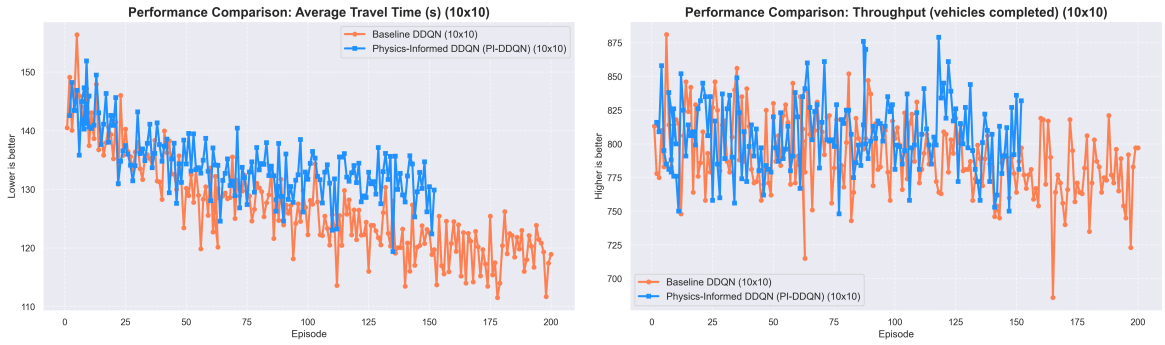


Figure 26: 10x10 DDQN vs PINN Travel Time and Throughput